

CrisisKD: Five-Stage Knowledge Distillation for Aspect-Level Sentiment and Emotion Analysis in Crisis Discourse

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Abstract

Identifying the target of emotional words or phrases in crisis situations, especially health-related ones, is important for understanding public concerns across cultural and linguistic contexts. We propose CrisisKD, a five-stage teacher–student knowledge distillation framework for aspect-level sentiment and emotion analysis on unannotated social media data. A teacher LLM generates aspect-level labels and reasoning traces that supervise a smaller student model across aspect extraction, syntactic parsing, opinion extraction, sentiment classification, and emotion classification. Using this framework, we construct and release a dataset containing 50,615 aspect-level labels, together with the annotation and fine-tuning scripts as open-source resources.¹ The resulting student supports end-to-end ABSA and emotion detection at substantially lower inference cost than the teacher. On a manually annotated 500-tweet gold set, the 5-task Qwen2.5-7B student improves over the untuned model by 7.9 F1 points on aspect extraction, 17.0 points on emotion accuracy, and 6.5 points on sentiment accuracy. On the external ABEA benchmark, CrisisKD improves the same-model Qwen2.5-7B ICL baseline by 2.8 F1 points on ATE and 3.8 F1 points on joint ATE+AEC.

Keywords: knowledge distillation; aspect-based sentiment and emotion classification; large language models; multi-step reasoning; crisis discourse; cross-domain generalization;

1 Introduction

During major health crises, such as the COVID-19 pandemic, public discourse becomes emotionally complex (Abd-Alrazaq et al., 2020). Understanding this discourse requires more than identifying whether a statement is positive or negative; we must also identify the specific topic, or “aspect,” targeted

by the emotion. For example, “fear” directed at the virus may imply willingness to cooperate with health measures, whereas “fear” directed at a vaccine may indicate reduced trust in its safety. This distinction is important because aggregate sentiment measures can obscure differences in public response (Pontiki et al., 2014). The same emotion may therefore signal different intentions depending on its target.

Aspect-Based Sentiment Analysis (ABSA) identifies topics and the sentiments expressed toward them, but standard approaches (Zhang et al., 2018) have largely focused on product reviews and coarse sentiment polarities rather than the fine-grained emotions, such as fear, anger, and hope, that shape behavior in crisis contexts. In this paper, we jointly model aspect extraction, aspect-based sentiment analysis, and aspect-level emotion detection on unannotated COVID-19 tweets.

A major barrier to aspect-level analysis is the lack of large COVID-19 datasets manually annotated for aspect-based sentiment and emotion classification (Banda et al., 2021). Existing datasets typically focus on named entities or sentence-level sentiment rather than aspect-level emotion categories. This limitation is particularly acute outside English, where annotated crisis data is even scarcer.

To reduce dependence on manual annotation, we propose CrisisKD, a five-stage teacher–student knowledge distillation framework. A large teacher model automatically annotates existing COVID-19 data and provides supervision for smaller student models. These students are more suitable for lower-resource deployment, and quantized variants can run on consumer hardware.

CrisisKD is informed by prior reasoning-based distillation work, including Syn-Chain (Fan et al., 2025), which shows that syntactic parsing and opinion extraction can improve ABSA knowledge distillation. Unlike approaches that assume existing aspect-level annotations, CrisisKD operates on raw

¹<https://github.com/MarkoHaralovic/CrisisKD>

text and generates both labels and reasoning traces for five tasks: aspect extraction, syntactic parsing, opinion extraction, sentiment classification, and emotion classification. The teacher outputs serve as supervision for the student, allowing it to learn both task labels and intermediate reasoning steps. Because the pipeline requires only raw text and a dependency parser, it is architecturally portable to other crisis domains and languages supported by suitable parsers such as spaCy (Honribal et al., 2020), although such transfer remains to be empirically validated.

1.1 Research Questions and Contributions

Our work addresses three questions:

- **RQ1:** Can LLM-generated synthetic annotations enable effective ABSA training on domain-specific data without human aspect-level annotations?
- **RQ2:** Do auxiliary syntactic-parsing and opinion-extraction tasks improve student performance on aspect extraction, sentiment classification, and emotion classification?
- **RQ3:** How do aspect-level emotions and sentiments differ in COVID-19 discourse, and what insights do they provide?

Our main contributions are: (1) a multi-step reasoning framework with explicit aspect extraction and emotion classification, architecturally adaptable to other crisis domains and languages; (2) an openly released COVID-19 ABSA dataset with emotion labels, annotation code, teacher prompts, and fine-tuned model weights; and (3) a student model that supports both downstream crisis-discourse analysis and automatic annotation of unannotated data, evaluated on a manually annotated gold set of 500 tweets.

2 Related Work

Weak Supervision and Automatic Labeling Interpreting event-specific tweets, such as COVID-19 discourse, is challenging because manually annotated data are scarce. Researchers therefore use **weak supervision**, where noisy or automatically generated labels provide training signals. Automatically generated pseudo-labels have proved viable for aspect-based sentiment analysis, particularly when syntactic structure is incorporated (Negi et al., 2024; Ratner et al., 2020).

Syntactic Dependency in ABSA Identifying the precise target of a sentiment requires modeling the grammatical relations that connect opinions to their targets. Prior work shows that syntax helps disentangle complex sentiments in English and Chinese COVID-19 discourse (Hou et al., 2025). Accordingly, CrisisKD includes syntactic parsing as an auxiliary task.

Emotion Detection in Crisis Discourse Fine-grained emotional targets, such as fear and anxiety, are important for understanding public health compliance and response (Abadi et al., 2021). Previous work trained downstream models from lexicon-based emotion annotations (Salsabila et al., 2023), but fixed lexicons struggle with context-dependent phenomena such as sarcasm and implicit emotion. CrisisKD instead uses contextualized reasoning from a large language model.

Reasoning-Based Knowledge Distillation Recent work moves beyond label prediction by transferring reasoning traces from a teacher model to a smaller student model (Fan et al., 2025). Such traces provide useful inductive bias for ABSA and can outperform label-only supervision. Building on this broader direction, CrisisKD provides a complete five-stage pipeline for unannotated data, jointly covering aspect extraction, syntactic parsing, opinion extraction, sentiment classification, and emotion classification.

Instruction Tuning Prompt formulation also influences ABSA performance. PFInstruct (Cabello and Akujuobi, 2024) shows that NLP-specific task prefixes improve results across ABSA subtasks. We follow this principle in the design of the CrisisKD prompts.

Cross-Lingual and Cross-Domain Portability Many ABSA and emotion systems remain restricted to English (Wu et al., 2025; Zhang et al., 2023). Multilingual parsers such as spaCy (Honribal et al., 2020) provide CoNLL-U representations for many languages, while multilingual COVID-19 datasets exist in Arabic, Spanish, and Chinese (Imran et al., 2022; Hou et al., 2025). Because CrisisKD requires only raw text, a dependency parser, and a multilingual teacher model such as Qwen2.5-32B (Yang et al., 2024), it is architecturally portable beyond English, although such transfer remains to be empirically validated.

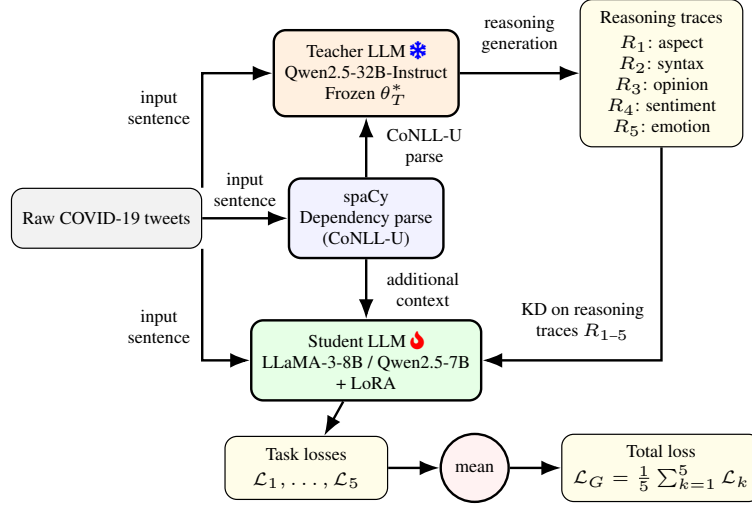


Figure 1: System overview of the proposed teacher–student framework. Each input sentence is parsed by spaCy into CoNLL-U format and, together with the raw sentence, is fed to a frozen teacher LLM (θ_T^*). The teacher produces multi-task reasoning traces R_1 – R_5 , which serve as knowledge-distillation targets for the student LLM (LLaMA-3-8B-Instruct or Qwen2.5-7B-Instruct, fine-tuned with LoRA). Task-specific generative losses $\mathcal{L}_1 \dots \mathcal{L}_5$ are averaged into $\mathcal{L}_G$.

3 Proposed Method

3.1 Method Overview

To address the challenge of identifying emotional targets without large-scale manual annotation, we propose CrisisKD, a teacher–student knowledge distillation framework visualized in Fig. 1.

CrisisKD uses the reasoning capabilities of a large language model (Teacher) to synthesize training data containing explicit reasoning traces that explain each label. These traces are then used to fine-tune a smaller, efficient Student model. This allows the Student to learn the reasoning required to identify why a sentiment or emotion is directed at a specific target while retaining the computational efficiency needed for large-scale analysis.

3.2 Data Selection

We selected COVID-19 tweet datasets that provide complete text for pseudo-annotation and knowledge distillation. Among available resources, including TweetsCOV19 (Dimitrov et al., 2020), COVIDSenti (Naseem et al., 2021), COVID19 NLP (Miglani, 2020), TB-COV (Imran et al., 2022), and METS-CoV (Zhou et al., 2022), many release only tweet IDs and therefore require Twitter/X API access. We consequently use **COVID-Senti** (Naseem et al., 2021), with approximately 90,000 tweets labeled as positive, negative, or neutral, and **COVID19 NLP** (Miglani, 2020), with 48,000 tweets labeled using five sentence-level sen-

timent categories. Neither dataset provides aspect-level sentiment or emotion labels.

We therefore pseudo-annotate both datasets with the teacher model, generating five reasoning traces (R_{1-5}) per sentence. These synthetic annotations are then used for knowledge distillation and student training. For cross-dataset evaluation, we additionally use the manually annotated **ABEA** dataset (Zorenböhmer et al., 2026), which provides aspect and emotion annotations independent of our pseudo-annotation pipeline.

Preprocessing Tweets lacking personal opinion or emotion, including headline-style posts, links, and quoted news, were removed using rule-based filtering. The remaining tweets were normalized, stripped of emojis and URLs, and parsed with spaCy’s (Honnibal et al., 2020) `en_core_web_sm` model into CoNLL-U format, with Unicode normalization performed using `ftfy` (Speer, 2019). Full filtering criteria and preprocessing details are provided in Appendix B.1.

3.3 Emotion Taxonomy

For aspect-level emotion classification, we adopt a 14-category emotion taxonomy designed for crisis discourse, together with one additional `no_emotion` fallback label for orphan annotations, failed parses, or cases where the teacher does not assign a valid emotion. The SenWave dataset (Yang et al., 2025) suggests the following seven categories: *optimistic*, *thankful*, *empathetic*,

pessimistic, anxious, sad, and annoyed. We extend this set with seven additional categories derived from fine-grained emotion taxonomies used in prior work (Demszky et al., 2020): *hopeful, proud, trustful, satisfied, scared, angry, and neutral*. We believe this taxonomy captures the details in emotional responses characteristic of pandemic discourse, distinguishing between related but distinct states such as *anxious* versus *scared*, or *optimistic* versus *hopeful*. The inclusion of *no_emotion* allows the model to handle factual statements about aspects where no affective content is present.

Unlike sentence-level emotion detection, our task assigns exactly one emotion label per aspect, enabling fine-grained analysis of how different targets within the same tweet evoke distinct emotional responses. For instance, a tweet may express *thankfulness* toward healthcare workers while simultaneously expressing *anxiousness* toward vaccine side effects.

This design reflects a social-psychological perspective on crisis emotions: distinct states such as *anxious* versus *scared*, or *optimistic* versus *hopeful*, map to different behavioral intentions and compliance patterns in public health contexts (Schröder et al., 2016). From the perspective of affect control theory, emotions are not merely internal states but socially and culturally shared sentiments that guide behavior (Schröder et al., 2016). Fine-grained aspect-level classification is therefore valuable beyond what coarse polarity models capture, particularly in multicultural crisis settings where the same emotional label may carry different behavioral implications across communities.

3.4 The Reasoning Pipeline

Our pipeline consists of five consecutive tasks.

Step 1: Aspect Extraction. Building on the syntactic structure, the model identifies aspects: specific entities, topics, or targets of opinion. For example, in “The vaccine rollout has been slow but the side effects are minimal,” the aspects are “vaccine rollout” and “side effects.” The teacher model generates a reasoning trace R_1 explaining why each span constitutes an aspect (e.g., “vaccine rollout” is identified because it represents a concrete policy action that opinions are expressed about). This task enables our student model to perform automatic aspect annotation.

Step 2: Syntactic Parsing. Dependency parses are obtained via spaCy (Honnibal et al., 2020) and

converted to CoNLL-U format, a tabular representation that explicitly encodes each word’s part-of-speech, head word, and dependency relation. This structured format is easier for LLMs to process than raw dependency trees (Fan et al., 2025; Matsuda et al., 2025). The teacher model generates a reasoning trace R_2 for syntactic parsing, learning to predict this structure from raw text. This serves two purposes: it provides syntactic features that inform the other tasks, and it acts as an auxiliary task that improves the student model’s linguistic understanding.

Step 3: Opinion Extraction. For each identified aspect, the teacher model extracts the opinion terms expressing sentiment toward that aspect. In our example, “slow” expresses opinion toward “vaccine rollout” and “minimal” toward “side effects.” Reasoning trace R_3 explains each opinion-aspect connection through dependency relationships, providing significant performance gains (Fan et al., 2025).

Step 4: Sentiment Classification. For each aspect-opinion pair, the model determines sentiment polarity (positive, negative, neutral). The reasoning trace R_4 explains why a particular sentiment label is appropriate given the opinion terms and broader context.

Step 5: Emotion Classification. For each aspect, the model identifies emotions expressed toward it (e.g., fear, anger, joy, trust, sadness). These categories are based on closed-set emotion taxonomies, which are relevant for crisis discourse. The reasoning trace R_5 explains the emotional content, which is more subtle than sentiment polarity.

Five reasoning traces were obtained from the teacher model Qwen2.5-32B-Instruct (Yang et al., 2024): R_1 for aspect extraction, R_2 for syntactic parsing, R_3 for opinion extraction, R_4 for sentiment classification, and R_5 for emotion extraction. The Qwen2.5-Instruct model family is reported to achieve state-of-the-art performance among open-weight models (Yang et al., 2024). Annotation of the corpus with Qwen2.5-32B-Instruct (Yang et al., 2024) on an NVIDIA L40 (48 GB) took approximately 7 days. The student models are Llama-3-8B (Grattafiori et al., 2024) and Qwen2.5-7B-Instruct (Yang et al., 2024), selected based on their strong distillation performance (Fan et al., 2025). Both models are fine-tuned using a LoRA adapter (Hu et al., 2022) via knowledge distillation from the teacher model’s reasoning traces. We

apply explicit teacher forcing between reasoning steps, i.e., step-level teacher annotations are provided as input rather than the student’s predictions from previous step. We adopt LoRA as an effective substitute due to its efficiency and the computational cost of full fine-tuning (Hu et al., 2022).

Loss definition The unified training objective across all five tasks is formulated as a generative loss:

$$\mathcal{L}_G = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \log P(g_{i,t} \mid \hat{g}_{i,<t}, C_i), \quad (1)$$

where N denotes the number of training samples, T is the length of the output sequence for each sample, $g_{i,t}$ is the ground-truth token at position t for sample i , $\hat{g}_{i,<t}$ is the previously generated token sequence up to position $t-1$, and $P(g_{i,t} \mid \hat{g}_{i,<t}, C_i)$ is the conditional probability of generating $g_{i,t}$ given the context C_i (which encodes the input sentence, aspect, and task-specific prompt) and the prefix $\hat{g}_{i,<t}$ (Fan et al., 2025).

Golden Set Evaluation The final fine-tuned model is evaluated against 500 manually annotated tweets derived from the COVID-Senti (Naseem et al., 2021) and COVID19-NLP (Miglani, 2020) datasets. Annotations followed a structured coding protocol. Inter-annotator agreement was assessed using span-level F1 scores for aspect extraction, and accuracy for sentiment and emotion labels on aligned aspect spans; disagreements were resolved through consensus. The finalized annotations constitute the gold standard used for evaluating model performance.

The main methodological contribution of CrisisKD is a five-stage distillation pipeline that jointly generates aspect, syntax, opinion, sentiment, and emotion supervision from raw text. This enables the student to operate both as a downstream ABSA system and as an automatic annotator for previously unannotated data.

3.5 Dataset Statistics

Teacher annotations were aggregated over the COVID19NLP (Miglani, 2020) and COVID-Senti (Naseem et al., 2021) datasets. The resulting corpus contains 21,937 sentences and 50,615 aspect instances, corresponding to an average of 2.307 aspects per sentence. Neutral sentiment is the most frequent label (51.87%), followed by negative (36.13%) and positive sentiment (12.01%). The most frequent emotion labels are neutral (42.70%),

annoyed (20.73%), and anxious (17.54%). Full label distributions and data-quality statistics are reported in Appendix B.2.

4 Experiments

We compare the performance of both student models on three tasks: aspect detection, sentiment classification, and emotion classification. Sentiment classification is a three-class problem (positive, negative, and neutral), while emotion classification uses 14 emotion categories and one additional `no_emotion` fallback label.

We validate the benefit of training on all five pipeline stages by removing components of the pipeline and evaluating the resulting model performance. This allows us to assess whether improvements in syntactic parsing or opinion extraction lead to improved downstream performance on the classification tasks.

We additionally compare model performance before and after fine-tuning to validate the efficacy of our approach. Teacher-model outputs are used as references, allowing us to measure alignment between teacher and student predictions. The model was trained with teacher forcing, whereas validation was performed end-to-end by propagating student responses from earlier tasks, thereby reproducing inference-time evaluation. We report task-level precision, recall, and macro F1, as well as accuracy for sentiment and emotion classification.

We evaluate CrisisKD in three settings: student-teacher fidelity, in-domain performance on a manually annotated gold set, and cross-dataset transfer on the external ABEA benchmark.

4.1 Training details

We use LoRA, a parameter-efficient fine-tuning method, where only a small subset of weights (adapter) is trained (Hu et al., 2022). For fair comparison between the two student models, Llama-3-8B-Instruct and Qwen2.5-7B-Instruct, we use the same fine-tuning recipe. Specifically, we use a learning rate of 2×10^{-5} , mixed-precision training, batch size of 8, and gradient accumulation of 8, resulting in an effective batch size of 64. We use an adapter rank of 16 targeting projection layers. For Qwen2.5-7B-Instruct we train 40M parameters ($\sim 0.5\%$), while for Llama-3-8B-Instruct we train 42M parameters ($\sim 0.5\%$ of total parameters). The dataset is split into 80% training and 20% validation sets, and the exact splits are released with

Table 1: Student–teacher fidelity across training scenarios. We report precision, recall, F1, and accuracy for sentiment and emotion classification, and precision, recall, and F1 for aspect extraction for each model against teacher outputs. Bold values indicate the best performance for a model, and * denotes the best performance overall.

Model	Scenario	Aspect Extraction			Emotion Cls.				Sentiment Cls.			
		Prec.	Rec.	F1	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1	Acc.
Llama-3-8B-Instruct	No fine-tuning	0.341	0.515	0.411	0.566	0.635	0.515	0.733	0.876	0.909	0.891	0.911
	Fine-tune (3 tasks, w/o opinions and syntax)	0.654	0.643	0.648	0.699	0.656	0.669	0.850	0.935*	0.912*	0.920*	0.929
	Fine-tune (4 tasks, w/o opinion extraction)	0.653	0.626	0.639	0.710	0.640	0.649	0.836	0.933	0.906	0.918	0.927
	Fine-tune (4 tasks, w/o syntactic parsing)	0.629	0.646	0.638	0.724*	0.657	0.675	0.852	0.933	0.899	0.915	0.926
	Fine-tune (5 tasks)	0.649	0.622	0.635	0.698	0.640	0.638	0.858	0.933	0.906	0.918	0.927
Qwen2.5-7B-Instruct	No fine-tuning	0.466	0.461	0.463	0.561	0.710*	0.587	0.791	0.903	0.905	0.903	0.922
	Fine-tune (3 tasks, w/o opinions and syntax)	0.638	0.658	0.648	0.699	0.682	0.670	0.852	0.930	0.900	0.915	0.929*
	Fine-tune (4 tasks, w/o opinion extraction)	0.696*	0.728	0.712	0.694	0.689	0.663	0.910	0.933	0.901	0.916	0.900
	Fine-tune (4 tasks, w/o syntactic parsing)	0.665	0.723	0.693	0.707	0.673	0.667	0.920	0.929	0.900	0.914	0.920
	Fine-tune (5 tasks)	0.681	0.747*	0.712*	0.710	0.696	0.703*	0.930*	0.923	0.901	0.912	0.910

the code. All student models were trained for 500 steps on a single NVIDIA L40 (48 GB), with training ranging 8-16 hours depending on the number of tasks in fine-tuning configuration.

4.2 Student–Teacher Fidelity Evaluation

Table 1 reports results for fine-tuning Llama-3-8B-Instruct and Qwen2.5-7B-Instruct. We evaluate four settings: (i) no fine-tuning, (ii) fine-tuning on downstream tasks only (3 tasks: aspect extraction, aspect-based sentiment classification, and emotion classification), (iii) fine-tuning on all five reasoning steps (adding syntactic parsing and opinion extraction), and (iv) ablations that omit one of the two additional reasoning tasks (syntax or opinion extraction) to test whether these traces improve student performance.

It is important to note that knowledge distillation in this setup is not fully model-agnostic: the token distribution and reasoning style are typically more similar within a model family than across families. Consequently, Qwen2.5-7B-Instruct may learn more effectively from the in-family teacher Qwen2.5-32B-Instruct than Llama-3-8B-Instruct.

Llama-3-8B-Instruct Results From Table 1, we draw the following conclusions for Llama-3-8B-Instruct: adding reasoning traces for syntactic parsing and opinion extraction does not provide meaningful benefit. The model achieves the best overall performance when fine-tuned only on the three downstream tasks (aspect extraction, sen-

timent classification, and emotion classification), reaching the highest aspect extraction F1 score of 64.8%, the highest sentiment classification accuracy of 92.9% across runs, and the second-highest emotion classification accuracy at 85.2%. Relative to the baseline, this corresponds to improvements of +23.5 pp in aspect extraction F1, +1.8 pp in sentiment classification accuracy, and +11.7 pp in emotion classification accuracy.

Qwen2.5-7B-Instruct Results For Qwen2.5-7B-Instruct, fine-tuning with reasoning traces yields substantial gains over the out-of-the-box baseline. The best overall configuration is obtained when fine-tuning on **all five tasks**, achieving the highest emotion classification accuracy (93.0%) and the strongest aspect extraction performance (F1 of 71.2%). Removing either syntactic parsing or opinion extraction slightly reduces overall performance, suggesting that the additional reasoning steps are beneficial for the Qwen student model. Relative to the baseline, the best model improves aspect extraction F1 by +24.9 pp, emotion classification accuracy by +13.9 pp, while sentiment classification, even though high 91.0%, drops by −1.2 pp compared to the baseline.

Student Results Comparison Comparing the best-performing configurations of both students, Llama (3 task fine-tuning) versus Qwen (5 task fine-tuning), we observe that the Qwen student consistently performs better on the downstream objectives. In particular, Qwen achieves higher as-

Table 2: Performance on the manually annotated gold set. We report accuracy and macro-F1 where applicable. Majority-class and uniform-random baselines are included for the classification tasks.

Task	Model	Acc.	F1
Aspect Extraction	Qwen2.5-7B Base	–	0.549
	Qwen2.5-7B FT (3)	–	0.583
	Qwen2.5-7B FT (5)	–	0.628
	<i>Teacher: Qwen2.5-32B</i>	–	0.709
Emotion Classification	Majority class	0.551	0.051
	Uniform random	0.305	0.077
	Qwen2.5-7B Base	0.508	0.569
	Qwen2.5-7B FT (3)	0.599	0.562
	Qwen2.5-7B FT (5)	0.678	0.570
	<i>Teacher: Qwen2.5-32B</i>	0.710	0.599
Sentiment Classification	Majority class	0.537	0.233
	Uniform random	0.423	0.333
	Qwen2.5-7B Base	0.658	0.692
	Qwen2.5-7B FT (3)	0.713	0.696
	Qwen2.5-7B FT (5)	0.723	0.696
	<i>Teacher: Qwen2.5-32B</i>	0.755	0.713

pect extraction performance (F1 score of 71.2% vs. 64.8%) and significantly higher emotion classification accuracy (93.0% vs. 85.2%), indicating better alignment with the teacher outputs when distilling within the same model family.

4.3 In-Domain Human Gold-Set Evaluation

Inter-coder agreement on the double-annotated subsets indicates generally consistent annotations across annotators. Aspect extraction achieves a mean F1 score of 73.0% across annotator pairs, while sentiment and emotion classification reach accuracies of 83.0% and 79.0%, respectively.

Table 2 reports results on the gold set of 500 tweets for the Qwen2.5-7B-Instruct baseline and our fine-tuned Qwen2.5-7B-Instruct students, trained on either 3 tasks or all 5 tasks in our pipeline, compared with the teacher model Qwen2.5-32B-Instruct. Fine-tuned models, as expected, outperform the baseline, with full fine-tuning across all 5 tasks showing better performance than 3-task fine-tuning. The performance of the fine-tuned model narrows the gap to the teacher model, demonstrating the effectiveness of our knowledge distillation framework.

4.4 Cross-Dataset ABEA Evaluation

Table 3 evaluates cross-dataset transfer to the ABEA benchmark. Compared with the same

Table 3: Performance on the ABEA test set. ATE denotes Aspect Term Extraction, AEC denotes Aspect Emotion Classification, and ICL denotes in-context learning. We compare our Qwen2.5-7B model against a same-size ICL baseline and a larger Qwen3-14B ICL model as an upper-bound reference.

Task	Model	Rec.	Prec.	F1
ATE	Qwen3-14B ICL	0.468	0.427	0.447
	Qwen2.5-7B ICL baseline	0.303	0.529	0.385
	Ours: Qwen2.5-7B	0.359	0.486	0.413
ATE + AEC	Qwen3-14B ICL	0.358	0.327	0.342
	Qwen2.5-7B ICL baseline	0.179	0.313	0.228
	Ours: Qwen2.5-7B	0.231	0.313	0.266

Qwen2.5-7B model used with in-context learning (ICL) our fine-tuned model improves ATE F1 from 0.385 to 0.413 and joint ATE+AEC F1 from 0.228 to 0.266. This indicates that our framework enables the 7B model to recover more relevant aspect and aspect–emotion instances than prompting the same model with ICL alone, narrowing the gap to the larger Qwen3-14B model.

4.5 Emotion–Aspect Associations via Enrichment Analysis

To better characterize fine-grained emotional structure beyond aggregate distributions, we analyze the association between aspects and emotions using a log-enrichment metric (Table 4). This measure points out aspects for which a given emotion is disproportionately expressed relative to its overall frequency in the dataset.

The results reveal that different aspects exhibit distinctive emotional signatures. Positive, socially oriented emotions (e.g., *proud*, *trustful*, *empathetic*) are strongly associated with human-centered aspects such as *doctor* and *people*, reflecting appreciation and collective solidarity. In contrast, negative emotions are concentrated around structurally different aspects: *angry* and *annoyed* are primarily linked to political and media-related aspects (e.g., *racism*, *trump*, *media*), while *pessimistic* and *anxious* are associated with broader systemic concerns such as *economy*, *pandemic*, and *travel*. Importantly, the same aspect appears across multiple emotion categories with different strengths. For example, *doctor* is strongly enriched for *proud*, *trustful*, and *empathetic*, while *shopping* is associated with both *optimistic* and *satisfied*. This shows aspect-level sentiment cannot be fully captured by coarse polarity labels, as distinct emotional inter-

Emotion	Top aspects (log-enrichment)
Proud	<i>doctor</i> (4.04), <i>people</i> (1.69), <i>vaccine</i> (1.48), <i>quarantine</i> (1.47), <i>shopping</i> (1.45)
Trustful	<i>doctor</i> (3.24), <i>hand sanitizer</i> (2.53), <i>World Health Organization</i> (2.52), <i>mask</i> (2.38), <i>government response</i> (1.89)
Optimistic	<i>shopping</i> (3.14), <i>health</i> (1.75), <i>cure</i> (1.74), <i>food</i> (1.68), <i>price</i> (1.66)
Thankful	<i>doctor</i> (3.13), <i>health</i> (2.48), <i>news</i> (2.26), <i>grocery/supermarket/store</i> (1.98), <i>shopping</i> (1.82)
Empathetic	<i>doctor</i> (3.06), <i>people</i> (2.43), <i>food</i> (2.15), <i>health</i> (2.10), <i>shopping</i> (1.78)
Hopeful	<i>cure</i> (3.04), <i>vaccine</i> (2.48), <i>news</i> (1.48), <i>shopping</i> (1.30), <i>world</i> (1.08)
Pessimistic	<i>economy</i> (3.03), <i>world</i> (2.69), <i>travel</i> (1.30), <i>italy</i> (1.15), <i>outbreak</i> (1.00)
Satisfied	<i>grocers response</i> (2.81), <i>price</i> (2.52), <i>shopping</i> (2.28), <i>food</i> (2.01), <i>flights</i> (1.73)
Sad	<i>doctor</i> (2.57), <i>deaths</i> (1.77), <i>quarantine</i> (1.19), <i>people</i> (1.12), <i>news</i> (0.92)
Angry	<i>racism</i> (2.50), <i>trump</i> (1.93), <i>public response</i> (1.84), <i>government response</i> (1.63), <i>government</i> (1.61)
No emotion	<i>vaccine</i> (2.39), <i>mask</i> (2.16), <i>doctor</i> (2.06), <i>shopping</i> (1.67), <i>government response</i> (0.80)
Annoyed	<i>media</i> (1.59), <i>panic buying</i> (1.33), <i>racism</i> (1.26), <i>trump</i> (1.24), <i>price</i> (1.03)
Anxious	<i>spread</i> (0.89), <i>travel</i> (0.83), <i>pandemic</i> (0.70), <i>economy</i> (0.65), <i>food</i> (0.40)
Neutral	<i>cruise ship</i> (0.58), <i>death toll</i> (0.58), <i>sars</i> (0.55), <i>wuhan</i> (0.55), <i>flights</i> (0.50)
Scared	<i>virus</i> (0.52), <i>covid/coronavirus</i> (0.26), <i>symptoms</i> (0.04), <i>panic</i> (-0.10), <i>travel</i> (-0.15)

Table 4: Top aspects per emotion ranked by log-enrichment, defined as $\log(P(\text{emotion} \mid \text{aspect})/P(\text{emotion}))$, over the top 50 aspects (minimum support $n \geq 200$). Positive values indicate that the emotion is more common for that aspect than in the dataset overall, negative values indicate underrepresentation, and 0 indicates baseline-level prevalence. Rows are ordered by the strongest top-ranked log-enrichment value in each emotion category.

pretations coexist within the same topic.

5 Discussion

With respect to the research questions introduced in Section 1.1, we organize the following sections to show how our findings support each question.

5.1 Effectiveness of Knowledge Distillation

Our results show that student models achieve substantial improvements across aspect extraction, sentiment and emotion classification. For Llama-3-8B-Instruct, the best configuration scores a +23.5 pp improvement in aspect extraction F1, a +1.8 pp improvement in sentiment classification accuracy, and a +11.7 pp improvement in emotion classification accuracy over the non-fine-tuned baseline. Similarly, Qwen2.5-7B-Instruct achieves large gains in aspect extraction (+24.9 pp F1) and emotion classification (+13.9 pp accuracy), providing evidence for the effectiveness of knowledge distillation with reasoning traces.

To assess the contribution of syntactic parsing and opinion extraction, we train models using all five tasks and ablate omitting one or both of these auxiliary tasks. For Llama-3-8B-Instruct, adding syntactic parsing and opinion extraction does not lead to additional gains. We attribute this to cross-architecture distribution mismatch, where reasoning traces generated by Qwen2.5-32B-Instruct may not align well with the internal representations of Llama-3-8B-Instruct, or where limited training ca-

capacity is spent modeling the teacher reasoning style rather than improving downstream performance.

In contrast, for the Qwen2.5-7B-Instruct student model, both syntactic parsing and opinion extraction provide measurable benefits. Fine-tuning on all five tasks results in the best overall performance, suggesting that the additional reasoning tasks are beneficial when teacher and student models belong to the same model family.

Overall, the results support claim that the proposed reasoning pipeline is effective. Exploring cross-architecture pairings and longer training schedules remains for future work.

5.2 Emotion Patterns in COVID-19 Discourse

Based on the teacher annotations, we observe that the majority of aspect-level sentiments are neutral (51.87%), and that 42.70% of aspect-level emotion labels are also neutral. At the sentence level, where annotations are available, we aimed to maintain similar label distributions.

It is possible for a sentence to exhibit an overall positive or negative valence while only the main aspect is assigned a corresponding sentiment/emotion label. Other, less central aspects that function as auxiliary nouns may be labeled as neutral because no explicit sentiment or emotional valence is expressed toward them. As a result, the distribution of neutral sentiment and emotion labels is skewed.

Another observation is the prevalence of negative sentiment labels, which occur approximately

three times more frequently than positive labels, and negatively associated emotions, which appear about five times more often than positive ones. This pattern reflects the general negativity present in pandemic discourse (Abadi et al., 2021; Lamsal, 2021). In crisis situations, such as COVID-19 pandemic, it is therefore particularly important to identify the specific targets of negative emotions, which highlights the value of our aspect-level analysis.

6 Conclusion

We propose CrisisKD, a five-stage teacher–student knowledge distillation pipeline for aspect-level sentiment and emotion analysis on unannotated COVID-19 tweets. The pipeline combines aspect extraction, syntactic parsing, opinion extraction, sentiment classification, and emotion classification, using a 14-category emotion taxonomy with an additional `no_emotion` fallback label. With Qwen2.5-32B as the teacher, we annotate 21,937 tweets with 50,615 aspect-level labels and fine-tune smaller student models, Llama-3-8B-Instruct and Qwen2.5-7B-Instruct, using LoRA. We evaluate the resulting models against both a manually annotated gold set of 500 tweets and the independently annotated ABEA benchmark. CrisisKD requires only raw text and a dependency parser, making it architecturally portable to other languages and crisis domains. Our contributions are:

- We propose CrisisKD, a five-stage distillation pipeline for aspect-level sentiment and emotion analysis on unannotated data.
- We construct and release a COVID-19 ABSA dataset with 50,615 annotated aspects across 21,937 tweets, together with annotation code and fine-tuned model weights.
- We demonstrate effectiveness on both human-annotated and external benchmarks. On a manually annotated 500-tweet gold set, the 5-task Qwen2.5-7B student improves over the untuned model by 7.9 F1 points in aspect extraction, 17.0 points in emotion accuracy, and 6.5 points in sentiment accuracy. On the external ABEA benchmark, it improves over the same-model ICL baseline by 2.8 F1 points on ATE and 3.8 F1 points on joint ATE+AEC.

Limitations

The framework relies on teacher-generated annotations, so teacher errors and biases may propagate

to the student models. Although the teacher generally produced well-formed outputs, its annotations should be treated as noisy supervision rather than ground truth. In particular, it occasionally generated semantically overlapping aspect spans; because we prioritized recall over precision, we did not apply additional aspect-merging or span-refinement beyond prompt tuning. The 14-category emotion taxonomy may also miss subtle or culturally dependent expressions, while annotation with Qwen2.5-32B remains computationally expensive.

Nevertheless, the distilled students support lower-cost inference, including deployment on consumer hardware through quantization. The pipeline can reduce the cost of constructing domain-specific ABSA resources, and its reasoning traces support error analysis and system refinement. Improved prompting, confidence-based filtering, or post-processing may further improve annotation precision. Although the method is architecturally portable to other domains and languages with suitable parsers, such transfer remains to be empirically validated.

Ethical Considerations

This work studies aspect-based sentiment and emotion analysis on publicly available COVID-19 Twitter datasets. The datasets used in this research, COVIDSenti (Naseem et al., 2021) and COVID19NLP (Miglan, 2020), consist of social media posts released for research purposes. No new human participant data were collected. Data was pseudo- and manually annotated, and annotations are released to support future research. The datasets do not contain direct personally identifiable information, reducing privacy risks, and the analysis focuses on collective discourse patterns rather than individual users.

However, analyzing emotional content in social media raises ethical concerns. Systems that identify sentiment or emotion may be misused for surveillance, monitoring without consent, or manipulative targeting. While this work aims to support crisis communication and public opinion analysis, similar methods could negatively affect privacy or autonomy. Practitioners should carefully evaluate such risks.

The gold standard dataset was annotated by four volunteer annotators from the research team, without financial compensation. Annotators were trained using a coding scheme defining emotion

and sentiment categories, examples, and rules for aspect spans, sarcasm, and edge cases. A calibration session was conducted before independent annotation. As annotators were team members performing an analytical task, this work did not undergo IRB review. No personal annotator data was collected, and participation was voluntary.

Bias and generalizability remain concerns. Social media data may not represent the full diversity of affected populations, and Twitter users are not demographically representative. Cultural differences also influence emotional expression. Additionally, the pseudo-annotation process relies on a large language model, which may introduce biases or errors. Thus, annotations should be interpreted as approximations of individuals' internal emotional states.

Although the proposed framework can be adapted to other languages and domains when raw text and linguistic analysis are available, the present study evaluates the approach only on COVID-19 Twitter data. The method assumes that aspect, sentiment, and emotion relations can be inferred from short-form text and syntactic structure; performance may degrade when these assumptions are violated (e.g., sarcasm, implicit context, or culturally dependent expressions). As a result, transfer to new settings remains to be validated with domain-specific data.

Finally, training large models has environmental implications. To mitigate this, the framework uses knowledge distillation and parameter-efficient finetuning (LoRA), reducing computational requirements, with only 0.5% of model parameters trained.

To support transparency and reproducibility, we aim to open-source training and evaluation code, along with derived annotation artifacts where licensing permits, and future work should independently assess dataset and method bias to ensure ethical use of the proposed approach.

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Appendix

A Task Logic and Example Annotations

A.1 Task Logic Visualization

Figure 2 presents the reasoning structure of the proposed framework.

Given a raw tweet, the model proceeds through five consecutive tasks: aspect extraction, syntactic parsing, opinion extraction, sentiment classification, and emotion classification. This decomposition follows prior work in aspect-based sentiment analysis and structured reasoning (Pontiki et al., 2014; Zhang et al., 2023; Fan et al., 2025), enabling the model to identify both targets and their associated sentiment and emotion.

The visualization emphasizes pipeline hierarchy. Aspect extraction determines the opinion targets, syntactic parsing provides grammatical structure, and opinion extraction links expressions to aspects. These intermediate representations then support aspect-level sentiment and emotion prediction. In our teacher-student setting, the teacher produces both labels and reasoning traces for each step, and the student is trained to imitate this full process, similar to recent structured reasoning approaches in ABSA (Fan et al., 2025).

A.2 Example Annotations

To illustrate the complete reasoning pipeline, we present representative examples that demonstrate our model’s capability to handle diverse COVID-19 discourse. These examples showcase the five-task reasoning chain (R1–R5) and highlight how the model disambiguates sentiment and emotion at the aspect level.

A.2.1 Handling Context

Figure 3 presents two contrasting cases that demonstrate the importance of aspect-level analysis.

The first example illustrates the model’s ability to correctly identify metaphorical usage of "Coronavirus" (referring to Corona beer rather than the virus), resulting in neutral sentiment and satisfied emotion despite surface-level ambiguity. The syntactic parsing (R2) and opinion extraction (R3) steps are crucial here, as they reveal the humorous context through dependency relationships.

In contrast, the second example shows genuine health concern, where incomplete grammar and informal language ("wish coronavirus not that serious") are correctly interpreted as expressing neg-

ative sentiment and anxiety toward the pandemic situation. These examples demonstrate how the reasoning traces enable the model to navigate linguistic complexity and contextual meaning that would challenge traditional sentiment analysis approaches.

A.2.2 Handling Different Emotions in Multi-Aspect Example

Figure 4 demonstrates the model’s capability on a more complex tweet containing two distinct evaluative targets.

The model successfully identifies both "media censorship" and "coronavirus outbreak" as separate aspects, performing independent reasoning chains for each. While both aspects receive negative sentiment classifications (R4), the emotion labels (R5) differ: anger toward censorship policies versus fear regarding outbreak severity. This distinction is semantically meaningful since anger reflects frustration with human decisions (censorship), while fear responds to the health threat itself.

The ability to extract and independently analyze multiple aspects within a single tweet is essential for understanding nuanced public discourse during crisis situations, where opinions often target multiple entities simultaneously with varying emotional valences.

A.2.3 Summary

These examples collectively demonstrate our model’s capabilities: handling informal social media language, disambiguating metaphorical and literal usage, attributing sentiments and identifying fine-grained emotions to individual aspects.

B Emotional and Sentiment Analysis of Teacher Annotations

B.1 Dataset Preprocessing

News Filtering A challenge using these datasets for aspect-based sentiment analysis is the number of "news headline" style tweets, such as factual statements, news links, and quoted headlines that lack personal opinion or emotion, and therefore introduce irrelevant noise during training. Such tweets are unsuitable for ABSA, as they do not express aspect-level sentiment. Tweets identified as news were removed during preprocessing. News tweets were detected using patterns such as URLs, common headline-style prefixes, source attributions, and truncated headline snippets.

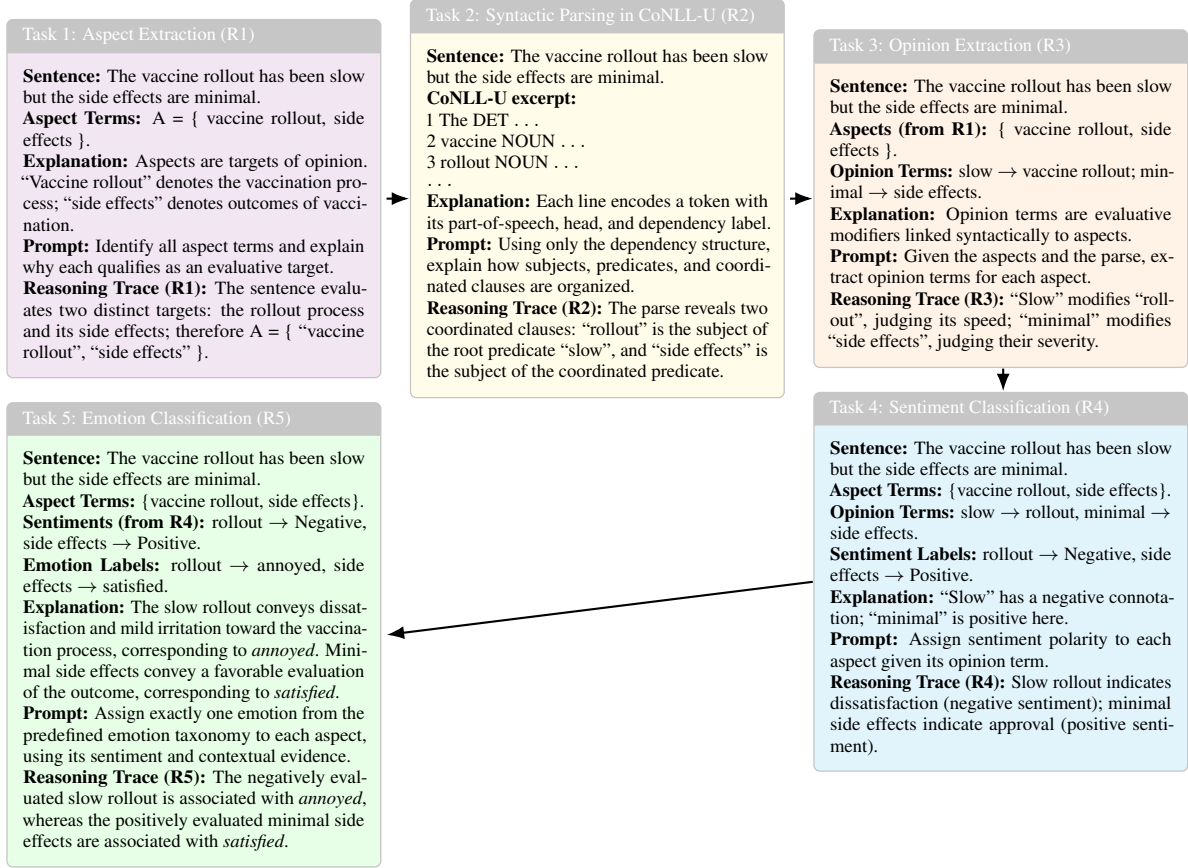


Figure 2: Reasoning example for aspect-based sentiment analysis and emotion detection on a single sentence via our teaching framework.

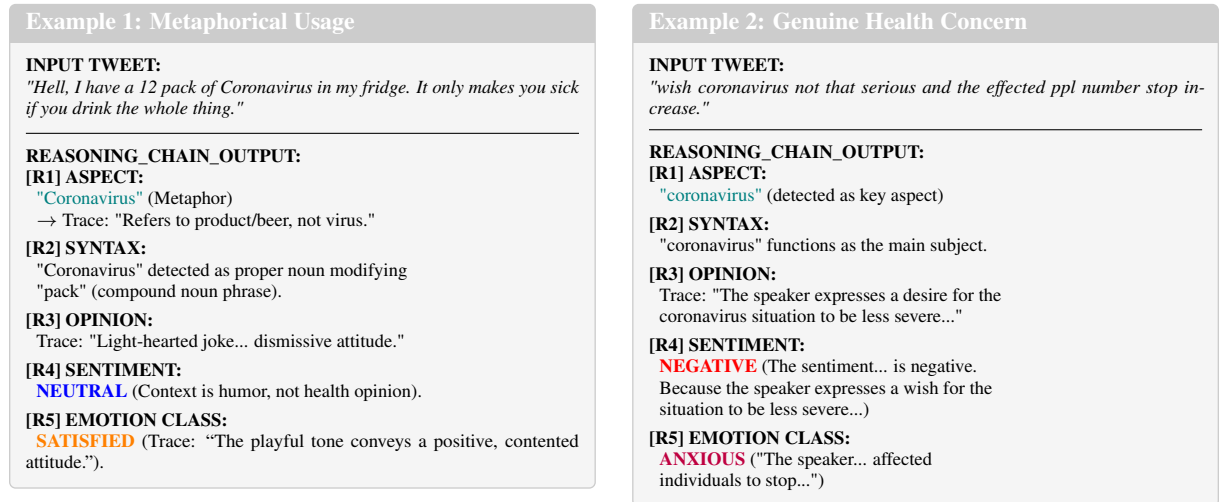


Figure 3: Example reasoning traces for two COVID-19 tweets demonstrating the complete annotation pipeline. Left: sarcastic reference requiring contextual sentiment disambiguation. Right: anxious expression of genuine health concern with negative sentiment.

Text Preprocessing The remaining tweets underwent additional preprocessing, including emoji removal, Unicode normalization using the `ftfy` (Speer, 2019) library, and URL stripping. We then performed syntactic pars-

ing using spaCy's (Honnibal et al., 2020) `en_core_web_sm` model, converting each tweet to CoNLL-U format, which encodes token-level information such as lemma, part-of-speech tags, and dependency relations.

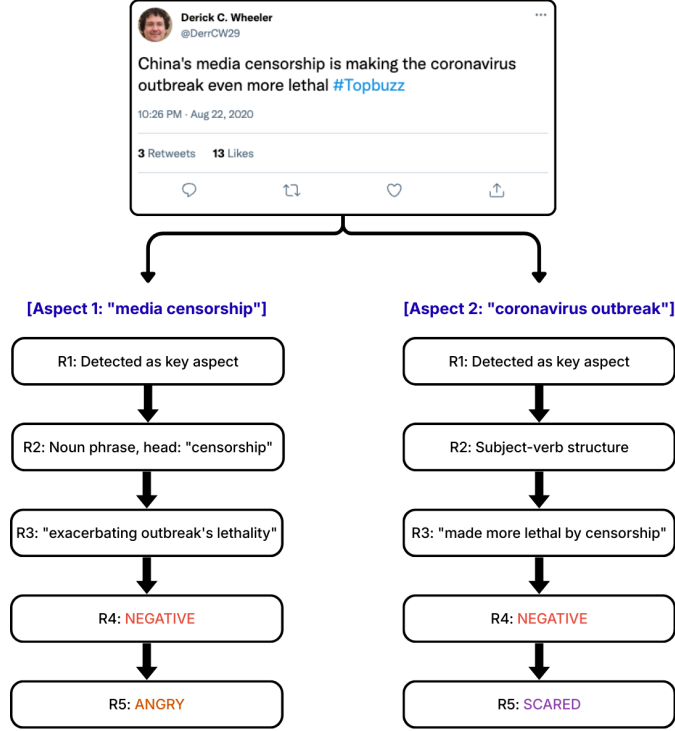


Figure 4: Illustrative reconstruction of a real tweet. Display name, profile image, username, timestamp, and engagement metadata are fictionalized. Reasoning trace across multiple aspects for a COVID-19 tweet. The model identifies two distinct aspects and performs the complete annotation pipeline for each, resulting in shared negative sentiment but divergent emotions: *angry* toward media censorship versus *scared* regarding the outbreak itself.

Table 5: Corpus summary based on Qwen2.5-32B annotations.

Metric	Value
Total sentences	21,937
Total aspects	50,615
Average aspects per sentence	2.307

Table 6: Aspect-level sentiment distribution ($N = 50,615$).

Sentiment	Count	Percent
Neutral	26,249	51.87%
Negative	18,288	36.13%
Positive	6,078	12.01%

B.2 Dataset Statistics

We report statistics computed from the Qwen2.5-32B (Yang et al., 2024) teacher annotations, aggregated over the COVID19NLP (Miglani, 2020) and COVIDSenti (Naseem et al., 2021) datasets.

The resulting corpus contains 21,937 sentences and 50,615 aspect instances, corresponding to an average of 2.307 aspects per sentence. During aggregation, we detected 13 orphan sentiment labels and 13 orphan emotion labels, defined as entries with missing aspect identifiers in the teacher model’s JSON output. These entries correspond to 0.026% of the aspect instances and were excluded from the training data and all subsequent analyses.

We analyze the distribution of emotion annotations across aspect categories to better understand the structure of public discourse captured by our

teacher model.

To improve interpretability, semantically equivalent aspect mentions (e.g., *coronavirus*, *covid*, *corona pandemic*, *covid crisis*) are normalized into unified aspect categories.

Figure 5 presents the emotion distributions for the 50 most frequent aspects. The results show that aspects related to pandemic are primarily associated with negative emotions, particularly anxiety and annoyance. In contrast, consumer-related aspects (e.g., *shopping*, *food*) exhibit more diverse emotional profiles.

The *vaccine* aspect is associated with hopeful emotion, reflecting expectations of recovery. In contrast, aspects related to political entities (e.g., *Trump*, *government response*) are largely associated with annoyance, indicating frustration toward

Table 7: Aspect-level emotion distribution ($N = 50,615$). The *no emotion* label indicates that the teacher did not assign an emotion to the aspect.

Emotion	Count	Percent
Neutral	21,608	42.70%
Annoyed	10,488	20.73%
Anxious	8,877	17.54%
Hopeful	1,640	3.24%
Scared	1,552	3.07%
Empathetic	1,436	2.84%
Satisfied	1,017	2.01%
Angry	1,014	2.00%
Optimistic	818	1.62%
Pessimistic	684	1.35%
Trustful	550	1.09%
Sad	449	0.89%
Thankful	262	0.52%
Proud	150	0.30%
No emotion	70	0.14%

decision-making and policy handling.

Overall, negative emotional valence dominates the dataset, with anxiety and annoyance being the most prevalent emotions, as public discourse during the pandemic is characterized by uncertainty and frustration. Aspects related to the pandemic often evoke anxiety due to perceived health risks and uncertainty, while secondary effects such as price increases, supply shortages, and quarantine measures contribute to annoyance and anger. Due to the large number of neutral tweets, where users write summaries or reports regarding certain aspects, neutral emotions account for a substantial proportion of each aspect’s emotion distribution.

C Human Annotation Set

The gold standard dataset was annotated by four volunteer annotators from the research team, without financial compensation. Annotators were trained using a coding scheme defining emotion and sentiment categories, examples, and rules for aspect spans, sarcasm, and edge cases. A calibration session was conducted before independent annotation.

In total, annotators labeled 500 tweets with intentionally overlapping assignments to enable agreement analysis. Annotator 1 labeled tweets 1–250, Annotator 2 labeled tweets 250–500, Annotator 3 labeled tweets 1–125 and 375–500, and Annotator 4 labeled tweets 125–375.

This design ensures that each subset of tweets is annotated by at least two annotators, allowing us to examine inter-coding agreement, following standard practices in dataset construction for senti-

Pair	Aspect (F1)	Sent.	Emo.
A1–A3	0.7493	0.8635	0.8229
A1–A4	0.7147	0.8289	0.7909
A2–A4	0.7386	0.7676	0.7042
A2–A3	0.7101	0.8516	0.8438
Mean	0.7282	0.8279	0.7905

Table 8: Inter-annotator agreement (aspect F1, sentiment accuracy, and emotion agreement).

ment and emotion analysis (Demszky et al., 2020; Naseem et al., 2021).

We report aspect agreement in F1 score, sentiment accuracy, and emotion accuracy between overlapping annotator pairs in Table 8.

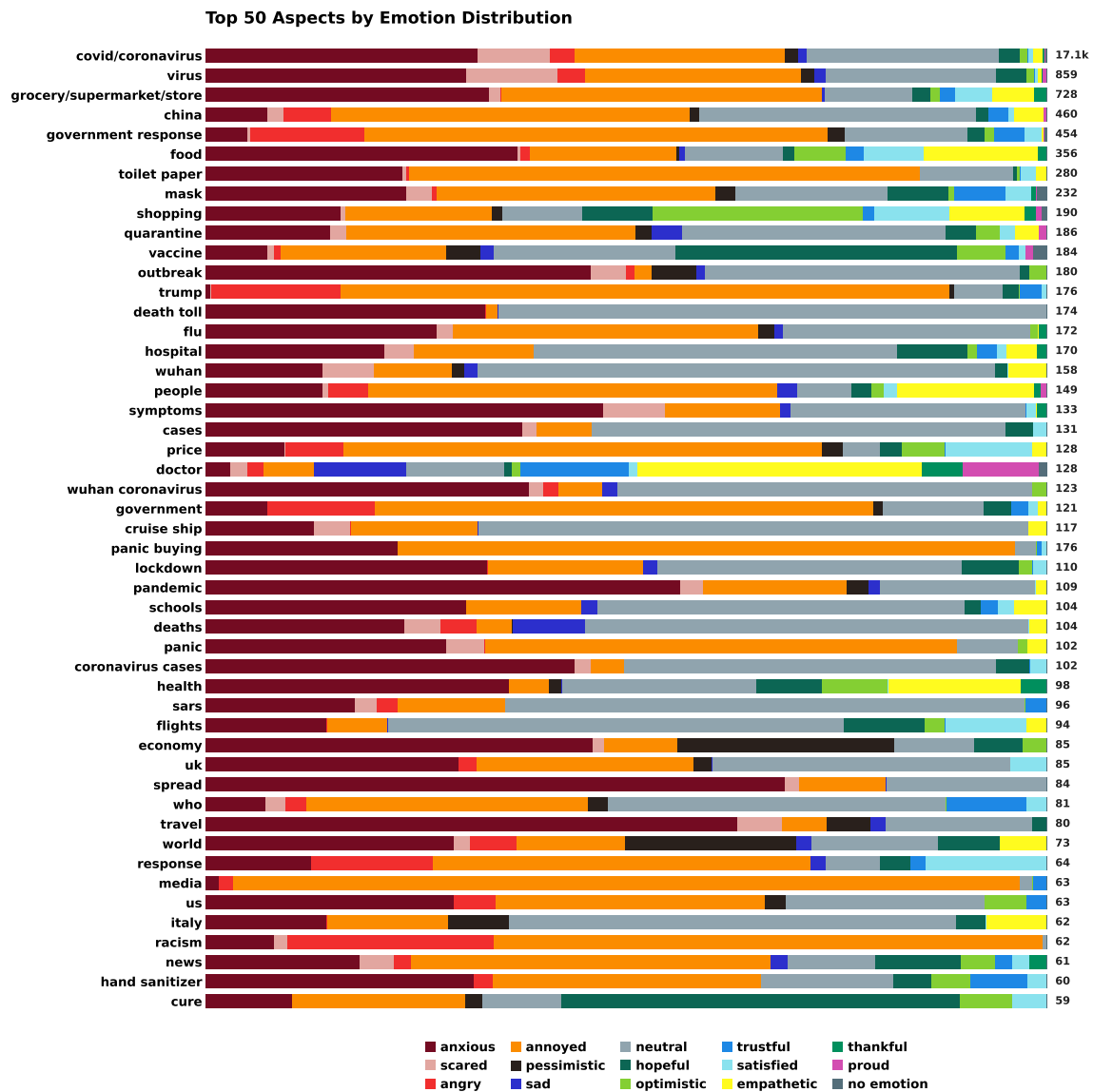


Figure 5: Emotion distribution across the 50 most frequent aspects. Each bar represents the normalized distribution of emotion categories for a given aspect.